

BUSINESS REQUIREMENTS DOCUMENT

Smart Student Productivity Platform

DOCUMENT CONTROL

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1. EXECUTIVE SUMMARY

The **Smart Student Productivity Platform** is a centralized, web-based academic productivity and management platform designed to bring essential academic activities into one integrated environment for Students, Faculty, Heads of Department (HOD), and authorized Administrators.

Students frequently depend on multiple disconnected platforms to manage classes, assignments, examinations, attendance, study materials, academic communication, online classes, personal tasks, and doubt resolution. Faculty members similarly use separate systems for assignment creation, evaluation, attendance, academic materials, communication, and online meetings. HODs and administrators have limited centralized visibility into departmental academic operations.

This fragmentation can result in:

- Scattered academic information
- Missed assignment and examination deadlines
- Inefficient communication
- Difficulty tracking academic performance
- Repetitive administrative work
- Limited visibility into faculty workload
- Delayed academic feedback
- Increased dependency on multiple external applications

The proposed platform addresses these challenges by providing a unified academic ecosystem containing student productivity tools, academic management, faculty workflows, departmental monitoring, communication, collaboration, and AI-assisted academic support.

The platform will provide separate role-based experiences for:

- Students
- Faculty
- HOD / Department Administrators
- Super Administrators

The solution will be implemented as a responsive web application using HTML5, CSS3 and JavaScript for the frontend and Firebase as the primary backend platform.

Firebase services will support authentication, database management, file storage, server-side functions, hosting, notifications and other required backend capabilities.

The platform will support academic structures such as:

Institution → Department → Program → Semester → Section → Subject → Faculty / Student

The BRD defines the business requirements, scope, stakeholders, business rules, feature requirements, technology direction, dependencies, risks, non-functional requirements and success criteria.

Detailed UI specifications, screen-level requirements, API specifications, database structures and implementation-level workflows will be maintained in the companion PRD and technical documentation.

2. REVISION HISTORY

Version	Date	Author	Description	Status
1.0	29 Aug 2026	BA	Initial BRD	Sign offed

3. INTRODUCTION

3.1 Purpose

This Business Requirements Document defines the business requirements for the Smart Student Productivity Platform.

The purpose of this document is to establish a common understanding between:

- Business stakeholders
- Product team
- Development team
- UI/UX team
- QA team
- Security team
- DevOps team
- Academic stakeholders

The BRD defines **what the platform must provide and why the capabilities are required**.

Detailed implementation specifications will be maintained separately in the PRD and technical design documentation.

3.2 Business Problem

Academic activities are often distributed across different systems and communication channels.

Students may use separate applications for:

- Timetables
- Assignments
- Study materials
- Attendance
- Examinations
- Results
- Communication
- Online classes
- Personal task management
- Doubt resolution

Faculty members may similarly use multiple systems to manage:

- Classes
- Students
- Assignments
- Assessments
- Attendance
- Learning materials
- Communication
- Online meetings

HODs require centralized visibility into departmental activities but may have to depend on manually collected information and multiple systems.

The absence of a unified platform creates operational inefficiencies and makes it difficult to obtain a complete view of academic activity.

3.3 Proposed Solution

The Smart Student Productivity Platform will provide a centralized web application that integrates academic management, student productivity, communication, collaboration and AI-assisted academic support.

The solution will provide:

Student Portal

Students will be able to:

- View academic information
- Access subjects
- View study materials
- Manage assignments
- Submit assignments
- View examinations
- Track attendance
- View results
- Manage tasks
- Use a unified academic calendar
- Communicate with faculty
- Join online classes
- Receive notifications
- Ask academic questions using AI
- Escalate unresolved doubts to faculty
- Create or join study rooms

Faculty Portal

Faculty will be able to:

- View assigned classes
- Manage subjects
- Upload learning materials
- Create assignments
- Review submissions
- Evaluate students
- Record attendance
- Manage assessments
- Schedule online classes
- Communicate with students

HOD Portal

HODs will be able to:

- Monitor departmental activities
- Allocate faculty
- Monitor student performance
- Monitor faculty workload
- Manage timetables
- Approve selected academic activities
- Generate departmental reports

Administration

Authorized administrators will manage:

- Users
- Roles
- Departments
- Programs
- Semesters
- Sections
- Subjects
- Academic configuration
- System settings
- Access permissions
- Audit information

4. BUSINESS OBJECTIVES

The platform shall pursue the following business objectives:

BO-01 — Centralize Academic Activities

Provide students and faculty with one centralized platform for core academic activities.

BO-02 — Improve Student Productivity

Help students manage academic responsibilities, deadlines, schedules and personal tasks.

BO-03 — Improve Academic Communication

Provide centralized communication between students, faculty and academic administrators.

BO-04 — Improve Faculty Efficiency

Reduce repetitive work associated with assignments, attendance, evaluation and communication.

BO-05 — Provide AI-Assisted Academic Support

Provide students with an AI-based first-level academic doubt resolution mechanism.

BO-06 — Improve Departmental Visibility

Provide HODs with centralized visibility into student performance, faculty allocation, workload and academic activities.

BO-07 — Support Multiple Academic Programs

Support configurable academic programs such as:

- B.Tech
- BBA
- MBA

The architecture should allow additional programs to be added without redesigning the platform.

BO-08 — Improve Data Accessibility

Ensure authorized users can access relevant academic information from a centralized system.

BO-09 — Improve Security

Protect academic records through authentication, authorization, role-based access and secure data handling.

BO-10 — Provide Scalable Infrastructure

Use Firebase services to provide managed backend infrastructure with minimal infrastructure-management overhead.

5. SCOPE

5.1 In Scope

The platform includes the following major areas:

Student Management

- Student dashboard
- Academic profile
- Subjects
- Courses
- Study resources
- Assignments
- Quizzes
- Projects
- Examinations
- Attendance
- Results
- Feedback
- Tasks
- Calendar
- Communication
- Online classes
- Notifications
- AI Doubt Solver
- Group Study Rooms
- Exam-focused study groups
- Profile and account security

Faculty Management

- Faculty dashboard
- Academic mapping
- Course management
- Study material management
- Assignment creation
- Assignment evaluation

- Attendance
- Assessments
- Examination evaluation
- Online meeting scheduling
- Faculty communication

HOD Management

- Department dashboard
- Faculty allocation
- Student monitoring
- Faculty monitoring
- Timetable management
- Academic approvals
- Department reports

Administration

- User management
- Academic structure management
- Role management
- Permission management
- System configuration
- Firebase service configuration
- Audit and security oversight

6. STAKEHOLDERS

Stakeholder	Responsibility / Interest
Product Owner	Owns product requirements and prioritization
Students	Primary users of academic and productivity features
Faculty	Manage academic delivery and student evaluation
HOD	Manage departmental operations and academic monitoring
Super Administrator	Manages platform-wide configuration

7. USER ROLES

7.1 Super Administrator

The Super Administrator has platform-wide administrative authority.

Responsibilities include:

- Managing administrators
- Configuring academic structures
- Managing roles
- Managing permissions
- Managing system settings
- Monitoring audit activities
- Managing platform configuration

7.2 HOD / Department Administrator

The HOD manages academic operations within the assigned department.

Responsibilities include:

- Managing faculty allocation
- Monitoring students
- Monitoring faculty
- Managing departmental timetable
- Reviewing academic information
- Managing academic approvals
- Generating departmental reports

7.3 Faculty

Faculty members manage academic activities for their assigned subjects and sections.

Responsibilities include:

- Managing classes
- Uploading materials
- Creating assignments
- Evaluating submissions
- Recording attendance
- Managing assessments
- Communicating with students

7.4 Student

Students use the platform to manage their academic activities and productivity.

Responsibilities include:

- Accessing academic information
- Viewing materials
- Completing assignments
- Tracking examinations
- Monitoring attendance
- Viewing results
- Managing tasks
- Asking academic questions
- Participating in study groups

8. ROLES VS. PORTAL ACCESS MATRIX

Capability	Super Admin	HOD	Faculty	Student
Student Portal	Support / Oversight	Department Oversight	—	Full Access
Faculty Portal	Support / Oversight	Department Oversight	Full Access	—
HOD Portal	Full / Support	Full Access	—	—
User Management	Full	Department Scope	—	—

Academic Structure	Full	Department Scope	View Assigned	View
Role Management	Full	Limited	—	—
Reports	Platform	Department	Assigned Classes	Personal
Audit Logs	Full	Department	Limited	Personal Activity
System Configuration	Full	—	—	—

9. BUSINESS REQUIREMENTS – FEATURE CATALOGUE

9.1 Student Portal

ID	Feature	Business Requirement	Priority
F01	Student Dashboard	The platform shall provide a personalized dashboard containing academic and productivity information relevant to the student.	High
F02	Academic Section	Students shall be able to view their academic structure and enrolled academic information.	High
F03	Courses / Subjects	Students shall be able to view enrolled subjects, faculty and relevant academic information.	High
F04	Study Resources	Students shall be able to access learning materials provided by authorized faculty.	High
F05	Assignment Management	Students shall be able to view, submit and track assignments.	High
F06	Quiz / Assessment	Students shall be able to participate in authorized quizzes and assessments.	Medium

F07	Projects	Students shall be able to track project requirements, milestones and submissions.	Medium
F08	Examinations	Students shall be able to view examination schedules, venues and instructions.	High
F09	Attendance	Students shall be able to view subject-wise attendance information.	High
F10	Results & Performance	Students shall be able to view published marks, grades and feedback.	High
F11	Productivity Tools	Students shall have access to personal productivity tools.	Medium
F12	Task Management	Students shall be able to create and manage academic and personal tasks.	Medium
F13	Calendar & Schedule	Students shall be able to view academic events, classes, examinations and deadlines in a unified calendar.	High
F14	Communication	Students shall be able to communicate with authorized faculty through the platform.	Medium
F15	Online Classes	Students shall be able to access scheduled online classes through integrated meeting links.	High
F16	Notifications	Students shall receive relevant academic notifications and alerts.	High
F17	AI Doubt Solver	Students shall be able to ask academic questions and receive AI-assisted explanations.	High
F18	Group Study Rooms	Students shall be able to create or join study groups with a maximum capacity of five students.	Medium
F19	Exam-Focused Study	Students shall be able to participate in exam-focused study groups.	Low
F20	Profile & Security	Students shall be able to manage permitted profile settings and account sessions.	Medium

9.2 Faculty Portal

ID	Feature	Business Requirement	Priority
F21	Faculty Dashboard	The platform shall provide faculty with a dashboard showing assigned academic activities.	High
F22	Academic Mapping	Faculty shall be able to view assigned subjects, sections and students.	High
F23	Course & Content Management	Faculty shall be able to upload and organize academic resources.	High
F24	Assignment Creation	Faculty shall be able to create, configure and publish assignments.	High
F25	Assignment Evaluation	Faculty shall be able to review submissions, provide marks and feedback.	High
F26	Attendance Recording	Faculty shall be able to record and manage student attendance.	High
F27	Assessment & Examination Evaluation	Faculty shall be able to evaluate assessments and publish authorized results.	High
F28	Online Meeting Scheduling	Faculty shall be able to schedule online academic meetings/classes.	Medium
F29	Faculty Communication	Faculty shall be able to communicate announcements and respond to student queries.	Medium

9.3 HOD Portal

ID	Feature	Business Requirement	Priority
F30	HOD Dashboard	HOD shall have a centralized departmental dashboard.	High
F31	Faculty Allocation	HOD shall be able to allocate faculty to subjects and sections.	High
F32	Student Monitoring	HOD shall be able to monitor student attendance and academic performance.	High

F33	Faculty Monitoring	HOD shall be able to monitor faculty allocation and workload.	Medium
F34	Timetable Management	HOD shall be able to manage departmental timetables.	High
F35	Academic Approval	HOD shall be able to approve authorized academic activities.	Medium
F36	Reports	HOD shall be able to generate departmental academic and operational reports.	Medium

9.4 ADMINISTRATIVE CAPABILITIES

ID	Capability	Business Requirement	Priority
AD-01	User Management	Authorized administrators shall create, update, deactivate and manage user accounts.	High
AD-02	Academic Structure Management	Administrators shall configure departments, programs, semesters, sections and subjects.	High
AD-03	Role & Permission Management	Administrators shall manage role-based access permissions.	High
AD-04	System Configuration	Authorized administrators shall manage platform-level configuration.	Medium
AD-05	Firebase Configuration	Authorized technical administrators shall manage required Firebase service configuration.	Medium
AD-06	Audit & Security Oversight	Authorized administrators shall review relevant audit and security events.	High

10. TECHNOLOGY & PLATFORM REQUIREMENTS

The platform shall use a Firebase-centered backend architecture.

Layer	Technology / Service
Frontend	HTML5, CSS3, JavaScript
Authentication	Firebase Authentication
Primary Database	Cloud Firestore
Real-Time Features	Firebase Realtime Database where required
File Storage	Cloudinary
Backend Logic	Firebase Cloud Functions
Hosting	Firebase Hosting
Notifications	Firebase Cloud Messaging
Security	Firebase Security Rules
Monitoring	Firebase / Google Cloud monitoring capabilities
Online Meetings	Google Meet integration
AI Backend Integration	Firebase Cloud Functions
Source Management	Project repository / version control system

11. BUSINESS RULES

ID	Business Rule
BR-01	The platform shall not provide unrestricted public self-registration.

BR-02	User accounts shall be created or provisioned by authorized administrators.
BR-03	A user shall access only information permitted by their assigned role and academic scope.
BR-04	Student access shall be restricted to the student's authorized academic context.
BR-05	Faculty access shall be restricted to assigned academic responsibilities.
BR-06	HOD access shall be restricted to the authorized department.
BR-07	Study rooms shall have a maximum capacity of five students.
BR-08	The system shall reject attempts to add a sixth member to a study room.
BR-09	Marks and results shall not become visible to students until they are officially published.
BR-10	Academic structure fields such as Program, Semester and Section shall be controlled by authorized administrators.
BR-11	AI requests shall be processed through a controlled backend mechanism.
BR-12	Deactivated users shall not be permitted to authenticate.
BR-13	Deactivated user records shall be retained where required for audit and institutional records.
BR-14	Sensitive academic data shall not be exposed through unauthorized APIs or client-side access.
BR-15	All critical administrative and academic actions shall be auditable.
BR-16	File uploads shall be validated before being made available to other users.
BR-17	Users shall not be able to modify information outside their authorization scope.

12. ASSUMPTIONS

The following assumptions apply to the current business requirements:

1. A Super Administrator will be provisioned during initial platform setup.
2. Authorized administrators will provision student and faculty accounts.
3. The institution will provide accurate academic master data.
4. Departments, programs, semesters, sections and subjects will be defined before operational use.
5. Firebase services will be provisioned before development and deployment.
6. Required Firebase billing configuration will be available when production usage requires paid services.
7. Google Meet access will be available for online classes where required.
8. Users will have internet connectivity to access the web platform.
9. Faculty and administrators will maintain accurate academic information.
10. AI functionality will be treated as an assistance mechanism and not as the authoritative source for academic decisions.

13. CONSTRAINTS

The platform will operate under the following constraints:

- The initial product will be a responsive web application.
- Native Android and iOS applications are not included in the initial scope.
- Firebase service quotas and pricing limits must be considered.
- Internet connectivity is required.
- Academic records depend on the accuracy of institution-provided data.
- AI responses may require human escalation.
- Google Meet availability is dependent on external service availability.
- Role-based access must be strictly enforced.
- File storage usage must be monitored as adoption increases.
- Firestore read/write optimization will be required for large-scale usage.

14. DEPENDENCIES

Dependency	Description
Firebase Project	Required for Authentication, Firestore, Storage, Functions and Hosting
Firebase Authentication	Required for secure user authentication
Cloud Firestore	Required for primary application data
Cloudinary	Required for academic files and submissions
Cloud Functions	Required for backend processing and AI integration
Firebase Hosting	Required for web application deployment
Firebase Cloud Messaging	Required for push notifications where applicable
Google Meet	Required for online class functionality
Academic Master Data	Required for initial platform configuration
Student Master Data	Required for student account provisioning
Faculty Master Data	Required for faculty account provisioning
AI Service Provider	Required for AI Doubt Solver functionality
Internet Connectivity	Required for platform access

15. RISKS & MITIGATIONS

Risk	Impact	Mitigation
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Firestore usage increases significantly	High	Monitor usage, optimize queries and establish appropriate billing limits
Unauthorized access to academic records	High	Implement Firestore Authentication and strict Security Rules
Incorrect academic master data	High	Validate and approve master data before import
AI provides inaccurate answers	High	Add AI guardrails and faculty escalation
Large file storage consumption	Medium	Enforce file limits and monitor Storage usage
Low user adoption	Medium	Provide onboarding and training
Google Meet unavailable	Medium	Provide alternate communication/rescheduling mechanisms
Poor network connectivity	Medium	Implement loading, retry and error states
Incorrect role assignment	High	Require administrator approval and validation
Database query inefficiency	Medium	Optimize Firestore data structures and queries
Notification delivery failure	Medium	Implement retry and fallback mechanisms
Academic structure changes	Medium	Provide controlled administrator workflows

16. NON-FUNCTIONAL REQUIREMENTS

16.1 Security

The platform shall:

- Authenticate users securely.
- Enforce role-based authorization.
- Protect academic records.
- Enforce Firestore Security Rules.

- Protect sensitive backend credentials.
 - Use HTTPS.
 - Maintain audit records for critical operations.
-

16.2 Performance

The platform should provide responsive user interactions under normal operating conditions.

The system shall optimize:

- Database queries
 - File loading
 - Dashboard data retrieval
 - API requests
 - Firebase reads and writes
 - Frontend assets
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16.3 Scalability

The solution shall support growth in:

- Students
- Faculty
- Departments
- Programs
- Courses
- Academic records
- Uploaded files
- Notifications
- AI requests

Firebase services shall be configured and optimized to support expected institutional growth.

16.4 Availability

The platform shall be designed to remain accessible during normal institutional operating periods and shall depend on Firebase service availability and configured service-level capabilities.

16.5 Usability

The platform shall provide:

- Clear navigation
 - Role-specific dashboards
 - Responsive layouts
 - Consistent UI patterns
 - Meaningful error messages
 - Loading states
 - Empty states
 - Confirmation messages
-

16.6 Accessibility

The platform should follow recognized accessibility principles, including:

- Keyboard navigation
 - Appropriate contrast
 - Descriptive labels
 - Accessible forms
 - Meaningful error messages
 - Responsive layouts
-

16.7 Reliability

The system shall handle expected failures such as:

- Network interruption
- File upload failure
- Notification failure
- Temporary backend unavailability
- Invalid input
- Unauthorized requests

16.8 Auditability

The platform shall record important actions including:

- User creation
 - User deactivation
 - Assignment creation
 - Assignment submission
 - Evaluation
 - Result publication
 - Attendance changes
 - Academic configuration changes
 - Permission changes
 - Administrative actions
-

17. SUCCESS CRITERIA

The project shall be considered successful when:

1. All approved MVP features are implemented and accepted.
 2. Students can manage core academic activities from one platform.
 3. Faculty can manage assignments, attendance, materials and evaluation.
 4. HODs can monitor departmental academic operations.
 5. Authorized administrators can manage users and academic structures.
 6. Firebase authentication and authorization operate correctly.
 7. Academic records are protected from unauthorized access.
 8. Students can receive AI-assisted academic support.
 9. Students can escalate unresolved doubts to faculty.
 10. Study rooms enforce the five-student capacity limit.
 11. Academic notifications are delivered for important events.
 12. The system provides reliable performance under expected institutional usage.
 13. Critical workflows are covered by QA testing.
 14. Production deployment can be performed through the defined Firebase deployment process.
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18. FEATURE-TO-OBJECTIVE TRACEABILITY

Feature ID	Feature	Primary Objective
F01	Student Dashboard	Centralize academic activities
F02	Academic Section	Centralize academic information
F03	Courses / Subjects	Centralize academic information
F04	Study Resources	Improve learning accessibility
F05	Assignment Management	Improve academic productivity
F06	Quiz / Assessment	Improve academic management
F07	Projects	Improve project tracking
F08	Examinations	Centralize examination information
F09	Attendance	Improve academic monitoring
F10	Results & Performance	Improve performance visibility
F11	Productivity Tools	Improve student productivity
F12	Task Management	Improve time management
F13	Calendar & Schedule	Improve planning
F14	Communication	Improve academic communication
F15	Online Classes	Support remote learning
F16	Notifications	Improve information delivery
F17	AI Doubt Solver	Provide academic assistance
F18	Group Study Rooms	Support collaboration
F19	Exam-Focused Study	Support examination preparation
F20	Profile & Security	Improve account management
F21	Faculty Dashboard	Improve faculty efficiency

F22	Academic Mapping	Improve academic organization
F23	Course & Content Management	Centralize learning resources
F24	Assignment Creation	Improve faculty efficiency
F25	Assignment Evaluation	Improve evaluation workflow
F26	Attendance Recording	Improve academic monitoring
F27	Assessment & Examination Evaluation	Improve evaluation
F28	Online Meeting Scheduling	Improve communication
F29	Faculty Communication	Improve collaboration
F30	HOD Dashboard	Improve departmental visibility
F31	Faculty Allocation	Improve resource allocation
F32	Student Monitoring	Improve student performance monitoring
F33	Faculty Monitoring	Improve faculty workload visibility
F34	Timetable Management	Improve academic planning
F35	Academic Approval	Improve governance
F36	Reports	Improve departmental decision-making

19. CHANGE CONTROL

Any significant change to the approved business requirements shall follow a controlled change-management process.

Step	Activity	Owner
1	Change Request	Requester
2	Business Impact Assessment	Product Owner

3	Technical Impact Assessment	Technical Lead
4	Cost and Timeline Assessment	Project Team
5	Approval	Business Sponsor / Product Owner
6	BRD Update	Product Team
7	PRD Update	Product Team
8	Backlog Update	Product Owner
9	Stakeholder Communication	Project Team

Changes may affect:

- Scope
- Features
- Business rules
- Firebase architecture
- Database structure
- Security requirements
- Timeline
- Cost
- Development effort

All approved changes shall be documented before implementation.

20. APPROVAL

The BRD shall be reviewed and approved by the relevant stakeholders before development proceeds against the approved business baseline.

Name	Title / Role	Signature	Date
Sayantan Ghosh	Client		

Sweta Kadam	Scrum Master — SR Pvt Ltd		
Sakshi Shingole	Project Manager + BA — SR Pvt Ltd		
Riddhi Zunjarrao	Tech Lead — SR Pvt Ltd		
Ragini Singh	Solution Architect — SR Pvt Ltd		